CUSTOMER 360° INTELLIGENCE

Customer 360° Intelligence & Churn Prevention Platform

End-to-End Data Analytics & Customer Retention Analysis

PYTHON | PANDAS | NUMPY | MYSQL | MONGODB | EXCEL | POWER BI | TABLEAU | ORANGE

Customer information is easier to use when the journey is connected

Customer profiles, orders, payments, returns, support, campaigns, and digital activity sit across business areas.

Viewed separately, they make the complete customer journey difficult to understand.

This project brings them together into one Customer 360° analytical view.

The project connects customer value, behaviour, engagement, and churn risk

01

Build a consolidated Customer 360° analytical dataset.

02

Understand purchasing behaviour and customer value.

03

Perform RFM analysis and identify churn-risk categories.

04

Analyse engagement and support-ticket activity.

05

Create interactive BI dashboards and practical retention-focused insights.

The final Customer 360° dataset combines 10,000 customers and 57 analytical features

Key fields include Customer Id, Customer Name, Customer Value Segment, RFM Segment, Total Revenue, Total Orders, Average Order Value, Days Since Last Order, Churn Risk Score, Churn Risk Category, Campaign Engagement, Digital Activity, and Support Ticket Status.

01
Raw Business Data

02
Data Validation

03
Data Processing

04
Customer 360° Dataset

05
RFM & Churn Analysis

06
BI Dashboards

07
Business Insights

Each technology had a clear role in the analytical workflow

Python

data generation and processing

Pandas

data manipulation and transformation

NumPy

numerical processing

MySQL

relational database and SQL analysis

MongoDB

event-oriented customer activity

Excel

PivotTables and business analysis

Power BI

interactive BI dashboard

Tableau

additional visualization

Orange

exploratory data analysis

Python turned separate processing stages into one repeatable workflow

01 Generate customer data
02 Generate sales/business data
03 Validate raw data
04 Load transactional data into MySQL
05 Generate MongoDB events
06 Build the Customer 360° dataset
07 Perform RFM analysis
08 Prepare the churn dataset
09 Prepare Power BI / Orange datasets
10 Validate the final analytical dataset

The project was organised into separate scripts for different processing stages.

MySQL handled relational business data; MongoDB added an event-oriented view

MySQL · novamart

- Customers, Products, Stores, Order Items, Payments, Returns, Support Tickets, and campaign-related data.
- Used for data loading, validation, and SQL analysis.

MongoDB

- Used for event-oriented customer activity.
- The Python workflow includes a MongoDB event-generation stage.
- It complements the relational data model without implying a production deployment.

Excel made the Customer 360° dataset easy to review from a business perspective

- Customers by Churn Risk Category
- Customer Value Segment Overview
- Revenue by Customer Value Segment
- Customer Distribution by RFM Segment
- Customer Churn Risk Distribution
- Customers by Support Ticket Status
- Executive Summary

PIVOTTABLE VIEW	CUSTOMERS
Critical	3,958
High	2,331
Medium	3,176
Low	525
Total	10,000

High + Critical = 6,289 customers in this PivotTable view.

These PivotTable values are not the same definition as the Power BI High Risk Rate metric.

Power BI brought the Customer 360° analysis into one completed report

COMPLETED REPORT

34

visuals across three pages.

Executive Overview

management-level customer and risk summary.

Churn & Retention

churn-risk and retention-focused analysis.

Customer 360° Profile

customer-level analytical view.

Tableau provided an additional analytical perspective

Customers by Churn Risk Category

Customers by Value Segment

Customers by RFM Segment

Average Churn Risk Score by Category

Customers by Digital Activity

Customers by Support Ticket Status

Tableau was used for additional visualization and analysis of the Customer 360° dataset.